

Experiment: Creating and Testing a Snapshot / Clone of a VM

Restoring a Previous Version using Oracle VirtualBox

AIM

To create a snapshot of a Virtual Machine (VM) in Oracle VirtualBox, make changes to the VM, and then test the snapshot mechanism by restoring the previous version. Additionally, to create a full clone of the VM and verify that it runs independently as a separate copy of the original.

THEORY

A **snapshot** is a saved state of a virtual machine at a particular point in time, including its disk data, memory state, and hardware configuration. Snapshots allow a VM to be reverted (rolled back) to an earlier state if something goes wrong after changes are made, without needing to reinstall or reconfigure the guest OS. VirtualBox stores snapshot data as differencing disk images layered on top of the base virtual disk. A **clone**, on the other hand, is a complete, independent copy of a VM. A **Full Clone** copies all data from the original VM into a new one with no ongoing dependency, while a **Linked Clone** shares the base disk image with the original and only stores the differences, saving disk space.

PROCEDURE

Part A: Creating and Restoring a Snapshot

- 1 Open **Oracle VirtualBox Manager** and select the target virtual machine from the list.
- 2 Start the VM and boot into the guest operating system to confirm it is functioning normally.
- 3 Make a note of the current state (e.g., create a test file on the desktop, or note the installed applications) so that changes can be verified later.
- 4 With the VM running (or powered off), go to **Machine > Take Snapshot** (or right-click the VM in the list and select **Snapshot > Take Snapshot**).
- 5 In the **Take Snapshot** dialog, enter a **Name** (e.g., "Before Changes") and an optional **Description**, then click **OK**. VirtualBox saves the current disk and memory state as a snapshot.
- 6 Resume working on the VM and make a noticeable change — for example, create a new file, delete the earlier test file, or install a small application.
- 7 Open the **Snapshots** tab in VirtualBox Manager to view the snapshot tree, which now shows the saved snapshot along with the **Current State** below it.
- 8 To test the snapshot, shut down the VM, right-click the snapshot named "Before Changes" in the Snapshots list, and select **Restore Snapshot**.
- 9 Confirm the restore action. VirtualBox reverts the VM's disk and state back to the exact point when the snapshot was taken.
- 10 Start the VM again and verify that the changes made after the snapshot (the new file, installed application, etc.) are no longer present, confirming the VM has been successfully rolled back to its previous version.

Part B: Creating and Testing a Clone

- 1 Power off the source VM completely (cloning generally requires the VM to be shut down).
- 2 Right-click the VM in VirtualBox Manager and select **Clone....**
- 3 In the **Clone Virtual Machine** wizard, enter a new **Name** for the cloned VM and choose the **MAC Address Policy** (usually "Generate new MAC addresses for all network adapters"), then click **Next**.
- 4 Select the **Clone Type**: choose **Full Clone** to create an entirely independent copy, or **Linked Clone** to create a copy that shares data with the original snapshot.
- 5 Choose the **Snapshot** to clone from — either the current state or a specific earlier snapshot (e.g., "Before Changes") — then click **Clone**.
- 6 VirtualBox copies the virtual disk and configuration files to create the new VM, which then appears as a separate entry in the VirtualBox Manager list.
- 7 Start the cloned VM and verify that it boots independently of the original and reflects the state of the snapshot/version it was cloned from.
- 8 Make a change in the cloned VM (e.g., create a file) and confirm that the original VM is unaffected, proving the clone operates as an independent copy.

Sample Observation Table

Stage	Action Performed	Observed Outcome
1	VM running, test file created on desktop	File present, VM functioning normally
2	Snapshot "Before Changes" taken	Snapshot listed in Snapshots tab
3	New file created / app installed	Current State shows changes
4	Snapshot restored	VM reverted to pre-change state
5	VM restarted after restore	Test file restored; later changes removed
6	Full clone created from snapshot	New independent VM appears in list
7	Clone booted and modified	Original VM remains unaffected

RESULT

A snapshot of the virtual machine was successfully created, capturing its state before any changes were made. After modifying the VM, the snapshot was restored and the VM correctly reverted to its earlier state, with all subsequent changes removed — confirming that VirtualBox's snapshot and restore mechanism works as expected. A clone of the VM was also created from the snapshot and verified to run as a fully independent virtual machine, unaffected by changes made to the original.

Conclusion: The experiment of creating a snapshot, restoring a previous VM version, and testing a cloned VM was performed successfully using Oracle VirtualBox.